

SE1 Module 3 Reviewer: Agile Methodologies

CS0025 Software Engineering 1, Module 3. Built from CS0025-M3S0, M3S1, and M3S2. Choice, matching, and typed lists.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Multiple Choice and True or False

1. Which statement best describes Agile software development?
 - A. Documentation must be completed before development begins
 - B. Customers participate only after the product is completed
 - C. Development follows a fixed sequence with no changes
 - D. Requirements and solutions evolve through collaboration

2. Agile is built on the collaborative effort of which kind of teams?
 - A. Separate departments that never meet
 - B. Only managers
 - C. Self-organizing and cross-functional teams and their customers or end users
 - D. Only external contractors

3. Which of these does Agile advocate?
 - A. Avoiding change requests
 - B. Documentation before any coding
 - C. Adaptive planning, evolutionary development, early delivery, and continual improvement
 - D. Fixed planning and one final release

4. Scrum is a type of methodology under which approach?
 - A. DSDM only
 - B. Crystal only
 - C. Agile
 - D. Waterfall

5. Which is one of the Agile Development Values?
 - A. Processes and tools over individuals and interactions
 - B. Contract negotiation over customer collaboration
 - C. Following a plan over responding to change
 - D. Working software over comprehensive documentation
6. Which Agile value says tools and processes are important, but competent people working together effectively matter more?
 - A. Individuals and interactions over processes and tools
 - B. Working software over comprehensive documentation
 - C. Customer collaboration over contract negotiation
 - D. Responding to change over following a plan
7. Which Agile value says a plan is important but must not be too rigid?
 - A. Customer collaboration over contract negotiation
 - B. Responding to change over following a plan
 - C. Working software over comprehensive documentation
 - D. Individuals and interactions over processes and tools
8. According to Agile principles, what is the primary measure of progress?
 - A. Number of meetings
 - B. Completed documentation
 - C. Project budget
 - D. Working software
9. Agile encourages teams to respond to changing requirements:
 - A. Only before development starts
 - B. Never after requirements are approved
 - C. Even late in development
 - D. Only during planning
10. According to the Agile principles, how often should working software be delivered?
 - A. Frequently, in weeks rather than months
 - B. Every few years
 - C. Once a year
 - D. Only at the end of the project

11. Which is the best form of communication according to the Agile principles?
- A. Contracts
 - B. Formal written reports
 - C. Face-to-face conversation (co-location)
 - D. Email only
12. Which Agile principle says the best architectures, requirements, and designs come from which kind of teams?
- A. Large management teams
 - B. Self-organizing teams
 - C. Outsourced teams
 - D. Teams with fixed roles only
13. Which Agile principle calls simplicity essential?
- A. Document everything
 - B. Do all possible work in advance
 - C. Add every requested feature
 - D. Simplicity is the art of maximizing the amount of work not done
14. Agile breaks product development work into what?
- A. Yearly releases
 - B. Separate models that never combine
 - C. Small increments that minimize up-front planning and design
 - D. One large phase
15. Which characteristic distinguishes Agile from the Waterfall model?
- A. Agile follows only one development phase
 - B. Agile discourages customer involvement
 - C. Agile uses an incremental and iterative approach
 - D. Agile requires all requirements to remain unchanged

16. In which model can the customer only see the product at the end of the project?

- A. Agile
- B. Waterfall
- C. Scrum
- D. Crystal

17. In which model is testing done only after the development phase?

- A. Waterfall
- B. Scrum
- C. XP
- D. Agile

18. How long are Agile iterations, according to the comparison table?

- A. One day
- B. Short, about 2 to 4 weeks
- C. Exactly one year
- D. 6 to 12 months

19. Which model gives documentation top priority?

- A. Scrum
- B. Agile
- C. Waterfall
- D. Lean

20. In Agile, how do testers and developers work?

- A. Together, with user acceptance at the end of every sprint
- B. Developers only
- C. Separately, with acceptance at the end of the project
- D. Testers only after release

21. Scrum is designed for teams of how many members?

- A. Ten or fewer
- B. At least twenty
- C. Exactly three
- D. Any size, with no limit

22. Scrum work is broken into timeboxed iterations called what?

- A. Cycles
- B. Sprints
- C. Releases
- D. Phases

23. What is the longest a sprint can be, and the most common length?

- A. One year; six months
- B. One day; one hour
- C. No longer than a month; two weeks is most common
- D. Three months; six weeks

24. Which is a benefit of Scrum listed in the module?

- A. Fewer team members
- B. No planning
- C. No customer feedback
- D. Flexibility and adaptability

25. Which Scrum role is the facilitator of the scrum development process?

- A. Stakeholder
- B. Product Owner
- C. Development Team
- D. Scrum Master

26. Which Scrum role represents the stakeholders and administers the scrum backlog?

- A. Tester
- B. Scrum Master
- C. Development Team
- D. Product Owner

27. Which Scrum role removes impediments to sprints and coaches and motivates the team?

- A. Customer
- B. Scrum Master
- C. Product Owner
- D. Development Team

28. How large is the scrum development team?

- A. Eleven to twenty
- B. Twenty or more
- C. One to two
- D. Three to ten individuals

29. Which Scrum role prioritizes goals for each sprint based on their value to stakeholders?

- A. Product Owner
- B. Tester
- C. Scrum Master
- D. Development Team

30. What does each sprint start with?

- A. A sprint review
- B. A release
- C. A retrospective
- D. A sprint planning event that establishes a sprint goal

31. How long is sprint planning for a 2-week sprint?

- A. Fifteen minutes
- B. One and a half hours
- C. Four hours
- D. Two hours

32. How long is the daily scrum?

- A. Four hours
- B. Two hours
- C. Fifteen minutes (timeboxed)
- D. One hour

33. Which is one of the three questions in the daily scrum?

- A. Which customer is unhappy?
- B. Do I see any impediment that could prevent me or the team from meeting our sprint goal?
- C. Who will be fired?
- D. What is the budget?

34. How long is the sprint review for a two-week sprint?

- A. Four hours
- B. Fifteen minutes
- C. Two hours
- D. One and a half hours

35. Which statement about the sprint review is correct?

- A. Incomplete work cannot be demonstrated
- B. It happens before the sprint starts
- C. Incomplete work must be demonstrated
- D. It is facilitated by the customer only

36. How long is the sprint retrospective for a two-week sprint?

- A. Four hours
- B. Fifteen minutes
- C. Two hours
- D. One and a half hours

37. Who facilitates the sprint retrospective?

- A. The Scrum Master
- B. The tester
- C. The customer
- D. The Product Owner

38. What is the backlog refinement formerly called?

- A. Retrospective
- B. Review
- C. Planning
- D. Grooming

39. Which Agile methodology focuses primarily on people and their interactions rather than processes and tools?

- A. FDD
- B. Crystal Methodology
- C. DSDM
- D. Waterfall

40. Who developed the Crystal Methodology?

- A. Ken Schwaber
- B. Kent Beck
- C. Mary Poppendieck
- D. Alistair Cockburn

41. Which sequence correctly represents the three phases of Crystal Methodology?

- A. Feasibility, Development, Maintenance
- B. Chartering, Cyclic Delivery, Wrap Up
- C. Planning, Coding, Testing
- D. Analysis, Design, Deployment

42. In Crystal, in which phase is a development team created and a preliminary feasibility analysis performed?

- A. Cyclic Delivery
- B. Wrap Up
- C. Deployment
- D. Chartering

43. In Crystal, in which phase does the integrated product get delivered to real users through delivery cycles?

- A. Chartering
- B. Feasibility
- C. Wrap Up
- D. Cyclic Delivery

44. In Crystal, in which phase are deployment into the user environment and post-deployment reviews performed?
- A. Cyclic Delivery
 - B. Wrap Up
 - C. Planning
 - D. Chartering
45. What does DSDM require of its users?
- A. Users must be actively involved, and teams are empowered to make decisions
 - B. Users stay away from the team
 - C. Users approve only at the end
 - D. Users write the code
46. Which DSDM technique completes a project incrementally by splitting it into portions, each with a fixed budget and delivery date?
- A. Chartering
 - B. Timeboxing
 - C. Prototyping
 - D. MoSCoW
47. In timeboxing, what happens to the requirements if time or money runs out?
- A. The lowest priority requirements are omitted
 - B. The budget is doubled
 - C. The deadline is extended
 - D. All requirements are dropped
48. What is MoSCoW used for?
- A. Testing
 - B. Scheduling meetings
 - C. Prioritizing requirements with stakeholders
 - D. Writing code

49. What does the M in MoSCoW stand for?

- A. Maintain
- B. Maybe
- C. Minimum
- D. Must have

50. What does the W in MoSCoW stand for?

- A. Won't have
- B. Wait
- C. Work
- D. Will have

51. Which DSDM technique creates early versions of the system so users can test-drive it?

- A. Chartering
- B. MoSCoW
- C. Prototyping
- D. Timeboxing

52. Which is the first phase of DSDM?

- A. Feasibility Study
- B. Implementation
- C. Pre-project
- D. Post-project

53. Which is the last phase of DSDM?

- A. Post-project
- B. Implementation
- C. Feasibility Study
- D. Pre-project

54. Feature Driven Development (FDD) is focused around what?

- A. Designing and building features
- B. Timeboxing
- C. Daily scrums
- D. Pair programming

55. Which is part of FDD's short phases of work per feature?

- A. Chartering
- B. Domain walkthrough, design inspection, promote to build, code inspection, design
- C. Story cardboard
- D. Sprint review

56. Lean software development is adapted from which system?

- A. The Agile Manifesto
- B. The Scrum guide
- C. The Waterfall model
- D. The Toyota Production System

57. Lean is based on which principle?

- A. Fixed scope, fixed time
- B. Just in time production
- C. Big bang release
- D. Documentation first

58. Which Lean principle says if an activity could be bypassed or the result achieved without it, it is waste?

- A. Eliminating Waste
- B. Amplifying Learning
- C. Defer Commitment
- D. Early Delivery

59. Which Lean principle says to delay decisions until they can be made on facts rather than assumptions?

- A. Building Integrity
- B. Defer Commitment
- C. Empowering the Team
- D. Early Delivery

60. Which Lean principle says that software value is measured in fitness for use, not conformance to requirements?
- A. Early Delivery
 - B. Amplifying Learning
 - C. Eliminating Waste
 - D. Optimize the Whole
61. Which Lean principle says the sooner the product is delivered without major defects, the sooner feedback comes?
- A. Building Integrity
 - B. Eliminating Waste
 - C. Early Delivery
 - D. Defer Commitment
62. Which Lean principle follows the Agile principle 'build projects around motivated individuals and trust them'?
- A. Empowering the Team
 - B. Defer Commitment
 - C. Optimize the Whole
 - D. Building Integrity
63. Which Lean principle refers to the customer's overall experience of the system?
- A. Amplifying Learning
 - B. Early Delivery
 - C. Eliminating Waste
 - D. Building Integrity
64. Which Lean phrase appears in the module?
- A. Think big, act small, fail fast; learn rapidly
 - B. Build big, ship late
 - C. Move slow, document everything
 - D. Plan once, build once

65. Extreme Programming (XP) is very helpful when what is true?

- A. Customer demands or requirements constantly change
- B. There is no customer
- C. Requirements never change
- D. The project has a fixed contract

66. XP advocates what kind of releases?

- A. No releases
- B. Frequent releases in short development cycles
- C. Only beta releases
- D. One yearly release

67. Which is the first XP phase in the module?

- A. Wrapping
- B. Execution
- C. Closure
- D. Planning

68. In which XP phase are stories captured in the 'Parking lot' and prioritized?

- A. Closure
- B. Analysis
- C. Planning
- D. Design

69. In which XP phase does coding and unit testing happen?

- A. Closure
- B. Wrapping
- C. Planning
- D. Execution

70. In which XP phase do pilot launch, training, and production launch happen?

- A. Closure
- B. Wrapping
- C. Planning
- D. Analysis

71. Which tool collects XP stories on sticky notes on a board?

- A. Story Cardboard
- B. Timebox
- C. Backlog refinement
- D. Online Storyboard only

72. Agile emphasizes individuals and interactions over processes and tools.

- A. True
- B. False

73. Agile considers comprehensive documentation more important than working software.

- A. True
- B. False

74. Scrum is a type of methodology under Agile.

- A. False
- B. True

75. Crystal was developed by Kent Beck.

- A. True
- B. False

76. A sprint review can demonstrate incomplete work.

- A. False
- B. True

Matching (Dropdown Style)

77. Match the Scrum item with its time or size.

- A. 2 hours
- B. 1.5 hours
- C. 30 minutes
- D. 4 hours
- E. 3 to 10 people
- F. 15 minutes
- G. 1 day

Daily scrum: _____

Sprint planning (2-week sprint): _____

Sprint review (2-week sprint): _____

Sprint retrospective (2-week sprint): _____

Scrum team: _____

78. Match the Agile method with its focus.

- A. People and their interactions
- B. Timeboxing and active user involvement
- C. Fixed requirements
- D. Designing and building features
- E. Eliminating waste
- F. Frequent releases in short cycles

Crystal: _____

DSDM: _____

FDD: _____

Lean: _____

XP: _____

79. Match the Lean principle with its description.

- A. Bypass any activity whose result can be achieved without it
- B. Trust motivated individuals to get the job done
- C. Delay decisions until they rest on facts
- D. Document everything first
- E. Deliver sooner to get feedback sooner
- F. Find and remove the root causes of defects

Eliminating Waste: _____

Defer Commitment: _____

Early Delivery: _____

Empowering the Team: _____

Optimize the Whole: _____

80. Match the Scrum role with its responsibility.

- A. Approves the budget
- B. Facilitates the process and removes impediments
- C. Self-organized group that does the actual work
- D. Represents stakeholders and administers the backlog

Scrum Master: _____

Product Owner: _____

Scrum team: _____

81. Match the model with its description.

- A. Only used for small hardware
- B. Incremental and iterative; customer sees the product early
- C. Flows sequentially; customer sees the product at the end

Agile: _____

Waterfall: _____

Typed Lists (Enumeration)

82. List the four Agile Development Values. (4 items)

1. _____
2. _____
3. _____
4. _____

83. List the three Scrum roles. (3 items)

1. _____
2. _____
3. _____

84. List the six Scrum events. (6 items)

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

85. List the three questions asked in the daily scrum. (3 items)

1. _____
2. _____
3. _____

86. List the three questions asked in the sprint retrospective. (3 items)

1. _____
2. _____
3. _____

87. List the three phases of Crystal Methodology. (3 items)

1. _____
2. _____
3. _____

88. List the seven phases of DSDM. (7 items)

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

89. List the four MoSCoW categories. (4 items)

1. _____
2. _____
3. _____
4. _____

90. List the seven principles of Lean software development. (7 items)

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

91. List the six phases of Extreme Programming (XP). (6 items)

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

92. List the four things that happen in each Crystal Cyclic Delivery cycle. (4 items)

1. _____
2. _____
3. _____
4. _____

93. List any four of the eight FDD practices. (4 items)

1. _____
2. _____
3. _____
4. _____

Answer Key

No.	Answer	Why
1	D. Requirements and solutions evolve through collaboration	Ch. 1: What is Agile
2	C. Self-organizing and cross-functional teams and their customers or end users	Ch. 1: What is Agile
3	C. Adaptive planning, evolutionary development, early delivery, and continual improvement	Ch. 1: What is Agile
4	C. Agile	Ch. 1: What is Agile
5	D. Working software over comprehensive documentation	Ch. 1: Agile values
6	A. Individuals and interactions over processes and tools	Ch. 1: Agile values
7	B. Responding to change over following a plan	Ch. 1: Agile values
8	D. Working software	Ch. 1: Agile principles
9	C. Even late in development	Ch. 1: Agile principles
10	A. Frequently, in weeks rather than months	Ch. 1: Agile principles
11	C. Face-to-face conversation (co-location)	Ch. 1: Agile principles
12	B. Self-organizing teams	Ch. 1: Agile principles
13	D. Simplicity is the art of maximizing the amount of work not done	Ch. 1: Agile principles
14	C. Small increments that minimize up-front planning and design	Ch. 1: What is Agile
15	C. Agile uses an incremental and iterative approach	Ch. 1: Agile vs Waterfall
16	B. Waterfall	Ch. 1: Agile vs Waterfall
17	A. Waterfall	Ch. 1: Agile vs Waterfall
18	B. Short, about 2 to 4 weeks	Ch. 1: Agile vs Waterfall
19	C. Waterfall	Ch. 1: Agile vs Waterfall
20	A. Together, with user acceptance at the end of every sprint	Ch. 1: Agile vs Waterfall
21	A. Ten or fewer	Ch. 1: Scrum
22	B. Sprints	Ch. 1: Scrum
23	C. No longer than a month; two weeks is most common	Ch. 1: Scrum
24	D. Flexibility and adaptability	Ch. 1: Scrum benefits
25	D. Scrum Master	Ch. 1: Scrum roles
26	D. Product Owner	Ch. 1: Scrum roles
27	B. Scrum Master	Ch. 1: Scrum roles

28	D. Three to ten individuals	Ch. 1: Scrum roles
29	A. Product Owner	Ch. 1: Scrum roles
30	D. A sprint planning event that establishes a sprint goal	Ch. 1: Scrum events
31	C. Four hours	Ch. 1: Sprint planning
32	C. Fifteen minutes (timeboxed)	Ch. 1: Daily scrum
33	B. Do I see any impediment that could prevent me or the team from meeting our sprint goal?	Ch. 1: Daily scrum
34	C. Two hours	Ch. 1: Sprint review
35	A. Incomplete work cannot be demonstrated	Ch. 1: Sprint review
36	D. One and a half hours	Ch. 1: Sprint retrospective
37	A. The Scrum Master	Ch. 1: Sprint retrospective
38	D. Grooming	Ch. 1: Backlog refinement
39	B. Crystal Methodology	Ch. 2: Crystal
40	D. Alistair Cockburn	Ch. 2: Crystal
41	B. Chartering, Cyclic Delivery, Wrap Up	Ch. 2: Crystal phases
42	D. Chartering	Ch. 2: Crystal phases
43	D. Cyclic Delivery	Ch. 2: Crystal phases
44	B. Wrap Up	Ch. 2: Crystal phases
45	A. Users must be actively involved, and teams are empowered to make decisions	Ch. 2: DSDM
46	B. Timeboxing	Ch. 2: DSDM techniques
47	A. The lowest priority requirements are omitted	Ch. 2: Timeboxing
48	C. Prioritizing requirements with stakeholders	Ch. 2: MoSCoW
49	D. Must have	Ch. 2: MoSCoW
50	A. Won't have	Ch. 2: MoSCoW
51	C. Prototyping	Ch. 2: Prototyping
52	C. Pre-project	Ch. 2: DSDM phases
53	A. Post-project	Ch. 2: DSDM phases
54	A. Designing and building features	Ch. 2: FDD
55	B. Domain walkthrough, design inspection, promote to build, code inspection, design	Ch. 2: FDD
56	D. The Toyota Production System	Ch. 2: Lean
57	B. Just in time production	Ch. 2: Lean
58	A. Eliminating Waste	Ch. 2: Lean principles
59	B. Defer Commitment	Ch. 2: Lean principles
60	B. Amplifying Learning	Ch. 2: Lean principles
61	C. Early Delivery	Ch. 2: Lean principles

62	A. Empowering the Team	Ch. 2: Lean principles
63	D. Building Integrity	Ch. 2: Lean principles
64	A. Think big, act small, fail fast; learn rapidly	Ch. 2: Lean
65	A. Customer demands or requirements constantly change	Ch. 2: XP
66	B. Frequent releases in short development cycles	Ch. 2: XP
67	D. Planning	Ch. 2: XP phases
68	B. Analysis	Ch. 2: XP phases
69	D. Execution	Ch. 2: XP phases
70	A. Closure	Ch. 2: XP phases
71	A. Story Cardboard	Ch. 2: XP tools
72	A. True	Ch. 1: Agile values
73	B. False	Ch. 1: Agile values
74	B. True	Ch. 1: What is Agile
75	B. False	Ch. 2: Crystal (Alistair Cockburn)
76	A. False	Ch. 1: Sprint review
77	Daily scrum = F; Sprint planning (2-week sprint) = D; Sprint review (2-week sprint) = A; Sprint retrospective (2-week sprint) = B; Scrum team = E	Ch. 1: Scrum events
78	Crystal = A; DSDM = B; FDD = D; Lean = E; XP = F	Ch. 2 methods
79	Eliminating Waste = A; Defer Commitment = C; Early Delivery = E; Empowering the Team = B; Optimize the Whole = F	Ch. 2: Lean principles
80	Scrum Master = B; Product Owner = D; Scrum team = C	Ch. 1: Scrum roles
81	Agile = B; Waterfall = C	Ch. 1: Agile vs Waterfall
82	Individuals and interactions over processes and tools, Working software over comprehensive documentation, Customer collaboration over contract negotiation, Responding to change over following a plan (any order)	Ch. 1: Agile values
83	Scrum Master, Product Owner, Scrum Development Team (any order)	Ch. 1: Scrum roles
84	Sprint; Sprint Planning; Daily Scrum; Sprint Review; Sprint Retrospective; Backlog Refinement (in order)	Ch. 1: Scrum events

85	What did I complete yesterday that contributed to the sprint goal?, What do I plan to complete today to contribute to the sprint goal?, Do I see any impediment that could prevent me or the team from meeting the sprint goal? (any order)	Ch. 1: Daily scrum
86	What went well during the sprint?, What did not go well?, What could be improved for better productivity in the next sprint? (any order)	Ch. 1: Sprint retrospective
87	Chartering; Cyclic Delivery; Wrap Up (in order)	Ch. 2: Crystal
88	Pre-project; Feasibility Study; Business Study; Functional Model Iteration; Design and Build Iteration; Implementation; Post-project (in order)	Ch. 2: DSDM phases
89	Must have; Should have; Could have; Won't have (in order)	Ch. 2: MoSCoW
90	Eliminating Waste; Amplifying Learning; Defer Commitment; Early Delivery; Empowering the Team; Building Integrity; Optimize the Whole (in order)	Ch. 2: Lean principles
91	Planning; Analysis; Design; Execution; Wrapping; Closure (in order)	Ch. 2: XP phases
92	Team updates and refines the release plan; Implements a subset of the requirements through one or more program test integrate iterations; Integrated product is delivered to real users; Review of the project plan and adopted development methodology (in order)	Ch. 2: Crystal cyclic delivery
93	Domain Object Modeling, Development by Feature, Component/Class Ownership, Feature Teams (any order)	Ch. 2: FDD (eight practices: also Inspections, Configuration Management, Regular Builds, Visibility of progress and results)